

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative modeler specializing in statistical model development and validation, hypothesis testing, regression, simulation, and model diagnostics. Experience evaluating financial-risk monitoring and insurance-pricing models, assessing data integrity and completeness, and working with large dependent datasets. Builds reproducible Python workflows and communicates model assumptions, performance, and limitations clearly.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected Dec 2026 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Platforms: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL/MySQL, R, Linux, Git, AWS EC2, HPC/SLURM

Model Development and Validation: Regression (GLM/GLS), predictive modeling, hypothesis and sequential testing, stochastic modeling, Monte Carlo simulation, numerical optimization, model selection and validation, performance assessment, data-quality and missing-data diagnostics

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Quantitative Finance / Risk Modeling

Summer 2026

- Developed and validated a Sequential Probability Ratio Test (SPRT) framework for binary risk indicators, translating a monitoring problem into explicit hypotheses and statistically controlled decision rules.
- Designed simulation studies to evaluate Type I/II error, detection delay, stopping-time uncertainty, and early-decision tradeoffs; documented model behavior, assumptions, and limitations.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pricing pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model validation, and performance diagnostics on AWS EC2.
- Evaluated model performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

MODEL DEVELOPMENT AND VALIDATION RESEARCH

Ph.D. Dissertation Research - Statistical Model Development and Diagnostics

Sep 2024 - Present

- **Model Development and Estimation:** Developed a scalable Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 observations per day.
- **Data Integrity and Completeness:** Built observation-mask, quality-filtering, nearest-neighbor, dependency-ordering, and irregular-grid workflows; quantified how missingness and resampling affect estimates.
- **Model Validation and Diagnostics:** Compared Matérn and generalized Cauchy specifications and designed scale- and frequency-resolved tests to identify misspecification in smoothness, noise, and dependence.
- **Quantitative Engineering:** Implemented reproducible Python/PyTorch pipelines with GLS profiling, L-BFGS optimization, CPU/GPU execution, cached artifacts, and restartable HPC jobs.

ADDITIONAL

Coursework: Econometrics, Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Microeconomics

Languages: Korean (Native); English (Fluent); Chinese (Basic)